

Natural Language Queries in Egocentric Videos

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Abstract

In this study, we explored the Ego4D dataset and its Natural Language Queries (NLQ) annotations to train VSLNet and its variants, VSLBase and VSLNet. These models were trained using features extracted through two approaches: Omnivore and EgoVLP. Additionally, we implemented an automatic query generation system leveraging Large Language Models (LLMs), supported by Gemma, to generate query-answer pairs from randomly sampled narrations.

Our experiments revealed that the BERT encoder outperformed VSLNet trained on SlowFast features in most cases. However, when using GloVe embeddings, notable improvements were observed only under specific configurations. Interestingly, during fine-tuning, the results reversed: GloVe embeddings outperformed BERT encoder, delivering unexpectedly strong performance. These findings highlight the intricate dynamics between feature extraction methods, model architectures, and embedding strategies in video-language tasks.

1. Introduction

Understanding actions and interactions in egocentric videos has emerged as a key area of focus in computer vision research, driven by the availability of large-scale datasets such as EPIC-KITCHENS [2] and Ego4D [7]. This project centers on the latter.

Ego4D is a large-scale, egocentric dataset of unparalleled diversity. The term "Ego" refers to egocentric perspectives, while "4D" signifies the inclusion of both 3D spatial and temporal information. The dataset comprises 3,670 hours of video captured by 931 unique participants from 74 locations across 9 countries. Data was recorded using seven commercially available head-mounted cameras, and portions of the dataset include additional modalities such as audio, 3D meshes of the environment, eye gaze, stereo video, and synchronized multi-camera recordings of the same event. Ego4D introduces five benchmark challenges designed to understand the first-person vi-

sual experience across three temporal dimensions: querying episodic memory (past), analyzing hand-object manipulations, audio-visual conversations, and social interactions (present), and forecasting activities (future). This work focuses on the episodic memory challenge, which involves forming, storing, and recalling conscious memories of specific past events alongside their contextual details. The goal is to analyze egocentric videos—captured via wearable cameras that record the "who/what/when/where" of an individual's daily experiences and, given a natural language query, localize the temporal window in the video where the answer can be found. For this project, we concentrate on one query type: Natural Language Queries (NLQ), where the query is expressed in text (e.g., "What did I put in the drawer?"), and the output is a temporal segment where the answer is visible or deducible.

In this study, we train and evaluate VSLNet on the NLQ benchmark using features extracted by Omnivore and EgoVLP. The performance of these models is compared to the baseline results provided in the Ego4D paper, which uses SlowFast 8x8 features with a ResNet-101 backbone pretrained on Kinetics-400 [3]. Furthermore, we introduce a novel variation of VSLNet by replacing its text encoder (BERT) with GloVe embeddings, examining how this modification affects the prediction of temporal segments.

In the final part of this work, we explore expanding the dataset. The current coverage of the Ego4D NLQ benchmark (227 hours) is limited compared to the full Ego4D dataset (3,600 hours). To address this gap, we propose augmenting the dataset by generating additional NLQ-like annotations using Large Language Models (LLMs), specifically Gemma. Our approach involves selecting sequences of consecutive video narrations and leveraging the LLM to generate synthetic query-answer pairs based on these narrations. The resulting temporal annotations are then used to train a new VSLNet model. Finally, we fine-tune this model using the original Ego4D NLQ dataset to enhance its performance.

2. Related Work

2D Temporal Adjacent Networks With Natural Language, it is an approach that focuses on the task of moment localization in videos, where the goal is to identify the precise start and timestamps of a moment based on a natural language query (*e.g.* “find the moment where the person opens the door”). The Moment Localization involves aligning a textual query with a specific segment in a video. The model leverages a 2D temporal representation to better capture the relationships between different temporal segments: instead of treating video frames or segments independently, the method models temporal relationships in a two-dimensional grid. Each grid point corresponds to a pair of timestamps (start and end), allowing the model to learn rich contextual information across adjacent temporal segments [15].

Natural Language Video Localization (NLVL) NLQ shares similarities with the Natural Language Video Localization (NLVL) task [14], where models must align a descriptive sentence to a specific temporal segment in a video. Key approaches in NLVL include TALL (Temporal Activity Localization via Language Query) [5], which models temporal context and semantic alignment, and 2D-TAN (2D Temporal Adjacent Networks), which considers adjacent temporal relationships for improved localization. However, NLQ poses additional challenges due to its question-answer format, requiring models to reason over both actions and objects with finer granularity.

Dataset Charades-STA is a dataset derived from the original Charades dataset by adding sentence-level temporal annotations. It consists of 9,848 videos depicting daily indoor activities, with 12,408 annotated sentence-temporal pairs. This dataset is designed for the task of temporal activity localization via natural language queries, enabling models to identify specific segments in videos corresponding to textual descriptions [5]. ActivityNet Captions is a large-scale benchmark for dense-captioning events in videos. It contains approximately 20,000 untrimmed videos sourced from YouTube, totaling around 849 hours of content. The dataset provides 100,000 temporally localized sentence descriptions, with each sentence uniquely aligned to a specific segment of a video. This resource facilitates research in both detecting and describing events within untrimmed video streams [8]. The TACoS (Temporally Anchored Commonsense Objective Scenes) dataset comprises videos focusing on cooking activities. It includes detailed temporal annotations linking natural language descriptions to specific video segments. The dataset is utilized for tasks such as temporal action localization and understanding the alignment between textual descriptions and corresponding visual content in cooking scenarios [12].

Data Augmentation with Large Language models

Data augmentation is a technique that leverages existing datasets to create additional training data without the need for new data collection [4]. This approach is particularly effective for enhancing model performance when data availability is limited [13]. In recent years, the use of Large Language Models (LLMs) to generate synthetic data for training has gained significant traction. Although LLMs can produce realistic and valuable training data, the success of this process depends on several factors, such as the choice of prompts, task complexity, and the quality, quantity, and diversity of the generated data [9].

Egocentric Video Understanding Recent advances leverage pretraining on large video-text datasets. Omnivore [9], a unified model for various visual modalities, and EgoVLP [10], tailored for egocentric vision with contrastive learning from natural language narrations, offer pre-extracted features. These models enable efficient training by reducing the computational load required for video-language tasks like NLQ.

- **Omnivore** model works on three different visual modalities: images, videos, and single-view 3D. It does not use a custom architecture for each visual modality. It performs recognition on all three modalities using the same, shared model parameters. It works by converting each input modality into embeddings of spatio-temporal patches, which are processed by exactly the same Transformer [8] to produce a representation of the input. It is trained to perform well across a variety of tasks, including those involving egocentric data, and provides a robust set of video features that can be leveraged for downstream tasks such as temporal segment prediction in Ego4D [6].
- **EgoVLP** (Egocentric Video-Language Pretraining) is a video-language model specifically designed for the Ego4D dataset. Trained using weak supervision from free-form textual narrations, EgoVLP provides high-quality video-language embeddings that are particularly useful for tasks like natural language queries (NLQ). These features capture the spatio-temporal dynamics of the video while aligning them with corresponding textual queries [10].
- **SlowFast Networks** for Video Recognition It is a state-of-the-art architecture for video recognition that addresses the need to capture both slow, coarse-grained motion and fast, fine-grained motion in videos. It achieves this by using two pathways with different frame rates [3]:

- slow pathway, that processes the video at a low

frame rate to focus on slow-moving, high level spatial information, such as understanding the overall context or scene structure.

- Fast pathway, that operates at a high frame rate to capture quick, detailed motion patterns, such as fast gestures or rapid movements in a video.
- Fusion of pathways: the two pathways are connected and exchange information, allowing the model to combine the strengths of both. The slow pathway provides context, while the fast pathway provides detailed temporal dynamics.

3. Methodology

3.1. Dataset

As presented before, we are using the EGO4D dataset, a comprehensive benchmark designed for various egocentric vision tasks. Starting with the annotations provided in the dataset, we conduct an in-depth analysis to understand how the queries are distributed across different dimensions. This step is crucial to ensure that the dataset is representative and provides sufficient coverage for training a robust model capable of handling diverse scenarios.

Distribution of queries (query size < 0.2) In histogram (Fig. 1) you can see a view of the distribution of short responses in terms of their relative query size (values less than 0.2). The x-axis represents the relative size of the query response. The y-axis shows the density, normalizing the frequency of occurrences to allow comparisons across different ranges. It is evident that the distribution is heavily right-skewed, with most responses concentrated towards the very low end of the query size range (close to 0). The highest density is observed in the smallest bin, highlighting that most responses are extremely short. As the relative query size increases, the density decreases significantly.

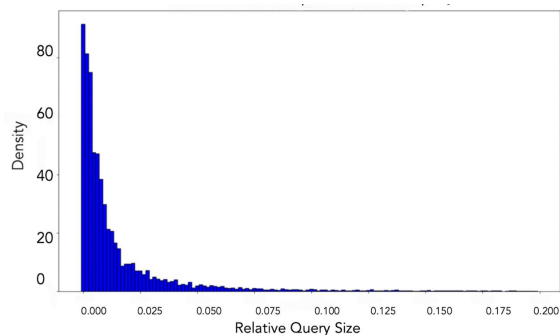


Figure 1. Distribution of queries (query size < 0.2)

Distribution of queries (query size > 0.2) This his-

togram (Fig. 2) shows the distribution of long responses based on their relative query size, where the size exceeds 0.2. The x-axis represents the relative query size, while the y-axis illustrates the density, which normalizes the frequency of occurrences across the range. From the picture, you observe a high density of responses clustered near the lower end of the range (close to 0.2). This suggests that even within the category of longer responses, most queries are relatively short. As the relative query size increases, the density gradually diminishes, indicating that very long responses are comparatively rare. This pattern could indicate that certain response lengths are more common due to specific characteristics of the dataset or the underlying tasks.

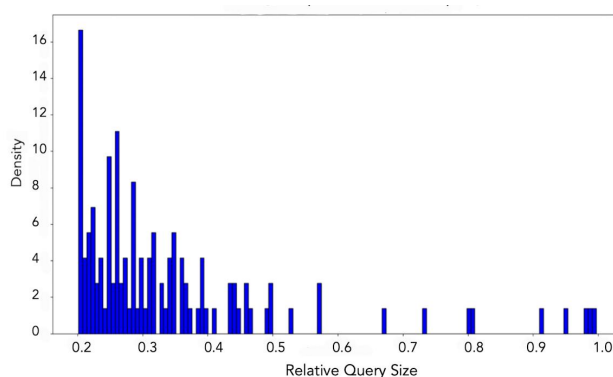


Figure 2. Distribution of queries (query size > 0.2)

Distribution (relative) of queries This bar chart (Fig. 3) illustrates the distribution of queries based on their template within the dataset. The x-axis represents different query templates, such as “Where is object X?”. Each of these templates corresponds to a distinct type of query, categorized by the nature of the question it addresses. The y-axis indicates the frequency, or the number of queries, associated with each template. From the chart, it is evident that certain templates are more common than others. These popular templates suggest that a substantial portion of the dataset involves locating objects or identifying their states. In contrast, templates that involve interactions with people, appear less frequently.

Clip sizes The histogram (Fig. 4) provides a visual representation of the distribution of video clip durations. Most of the clips are concentrated around a specific duration range, which is evident from the tall spike near 400 seconds. Interestingly, there is another smaller but noticeable spike at the far end of the chart, near 1200 seconds. This could imply that the dataset consists of mostly standard-length clips, with a few extended ones included.

Distribution of the answer segments along the videos The histogram (Fig. 5) provides a clear picture of when responses typically occur during the course of the videos.

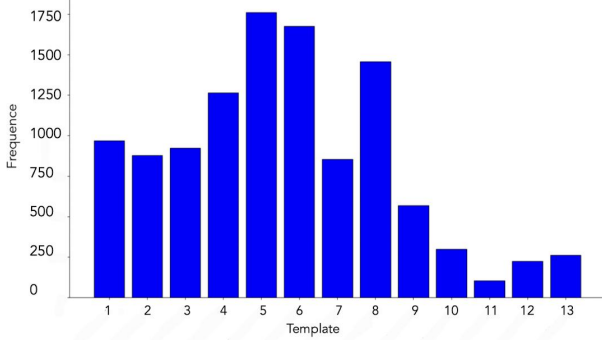


Figure 3. Distribution (relative) of queries according to their template

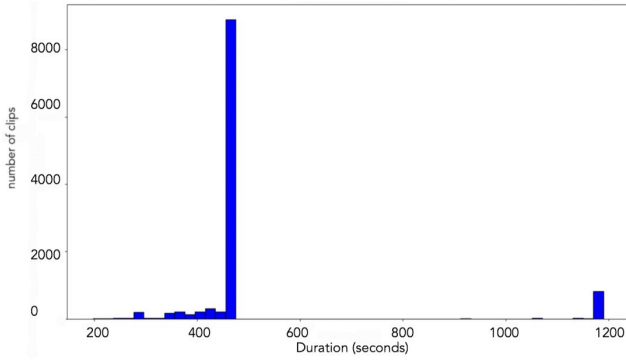


Figure 4. Clip Sizes

Most responses start very early in the videos, as shown by the prominent peak near the beginning of the timeline. This indicates that the majority of interactions or key moments happen shortly after the videos begin.

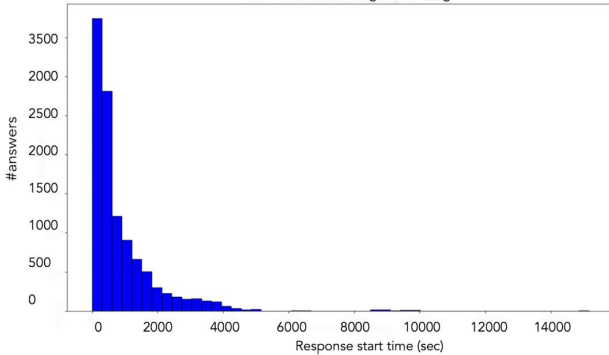


Figure 5. Distribution of the answer segments along the videos

3.2. Architecture

The proposed architecture tackles Natural Language Video Localization (NLVL) using a span-based question answering framework enhanced by Query-Guided High-

lighting (QGH). It consists of multiple interconnected blocks, including feature extraction, feature encoding, attention mechanisms, and span prediction, all working together to accurately predict the correct temporal span in response to a given query.

Feature Extractor The Feature Extractor block processes both video frames and text inputs to generate meaningful representations. For video processing, the block employs C3D (3D Convolutional Neural Networks), which applies three-dimensional convolutions across video frames to simultaneously capture spatial and temporal features. The C3D model, pretrained on the large-scale Kinetics dataset [1], generates robust feature representations without requiring additional training. For text processing, the block utilizes the established GloVe (Global Vectors for Word Representation) framework [11]. The Feature Extractor produces two distinct output structures: a sequence of visual feature vectors $V = \{v_1, v_2, \dots, v_n\}$ representing the video content, and query embeddings $Q = \{q_1, q_2, \dots, q_n\}$ for the text components. These representations capture the semantic content and temporal relationships within both modalities, enabling downstream processing of the multimodal information.

VLSBase and VLSNet VLSNet [2] is an enhancement of its predecessor, VLSBase, which does not fully address the key differences between video and natural language processing. While VLSNet shares the same basic components and overall flow as VLSBase, our focus is on analyzing the improved version due to its advancements.

VLSNet takes features from the previous block and fed into a Feature Encoder throw a feed-forward networks which is composed of a normalization layer and residual connection for each layer. FE is a single encoder block that aligns them in a shared vector space through four convolutional layers and a multi-head attention. To capture cross-modal interactions, the encoded video and query features are passed through the Context-Query Attention (CQA) module, which captures cross-modal interactions. This module first calculates similarity scores and then derives context-to-query (A) and query-to-context (B) attention weights, producing a query-aware representation of the video.

The Query-Guided Highlighting (QGH) module refines this representation by focusing on the most relevant video regions (foreground) while ignoring irrelevant regions (background). It extends the boundaries of the foreground region to include its antecedent and consequent, assigning a binary classification label of 1 for foreground and 0 for background. This binary classification task predicts the confidence of whether a visual feature belongs to the foreground or background. The loss for the QGH module

(Eq. (1)) is computed using a binary cross-entropy function L_{QGH} , which compares the predicted highlighting scores S_h with the ground truth labels Y_h :

$$L_{QGH} = f_{CE}(S_h, Y_h) \quad (1)$$

Finally, a Conditioned Span Predictor uses two unidirectional LSTMs and two feed-forward to predict the start and end boundaries of the temporal span corresponding to the query, enabling the model to precisely identify the target segment. The span prediction loss L_{span} (Eq. (2)), also computed using a binary cross-entropy loss function, is defined as:

$$L_{span} = \frac{1}{2} [f_{CE}(P_s, Y_s) + f_{CE}(P_e, Y_e)] \quad (2)$$

Where:

- P_s and P_e : Predicted probabilities for the start and end boundaries.
- Y_s and Y_e : Ground truth labels for the start and end boundaries.

The overall loss (Eq. (3)) combines the QGH loss (L_{QGH}) and the span prediction loss (L_{span}) for start and end boundaries to train the network end-to-end.

$$L = L_{span} + L_{QGH} \quad (3)$$

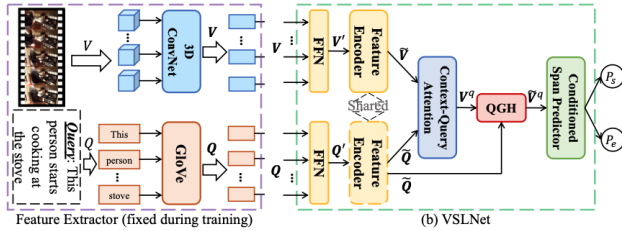


Figure 6. Structure of the VLSNet

3.3. FineTuning

The primary objective is to enhance the model by incorporating additional data, utilizing Gemma as a key tool. The process to expand the number of queries begins with the annotations file in the EGO4D dataset. Unlike previous work, which focused exclusively on the NLQ data, we now exclude this subset and concentrate on the remaining portions of the dataset.

To maintain consistency, we adhered to the same principles used in the previous dataset, preserving the file structure. Annotations were divided into segments of N consecutive narrations, creating larger narration blocks. The

narrations were segmented based on the structure of the initial dataset, dividing them into 10-second intervals. This was achieved through a granular analysis of each video, calculating the average number of narrations required to align with the 10-second timeframe. To define each group, the starting point was set as the initial timestamp of the first narrative, while the endpoint corresponded to the initial timestamp of the subsequent video. Although this method provides a systematic approach, it inherently introduces a greater degree of inaccuracy compared to alternative strategies.

Subsequently, we utilized Gemma, a large language model (LLM) to generate the required queries. For each generation, the number of tokens was dynamically calculated by multiplying the number of selected narrations by a constant value of 5, which was optimized to reduce errors. A maximum token limit of 1024 was imposed. After applying necessary refinements and adjustments, the dataset was finalized and prepared for submission to VLSNet for pretraining. Finally, the model underwent fine-tuning by using the pretrained model as a parameter to optimize performance.

4. Experimental Results

A crucial step in training models like VLSNet is the extraction of visual features from the raw video data. Given the computational challenges of training video models from scratch, pre-extracted visual features have become a popular and efficient solution. For this project, the Omnivore [9] and EgoVLP [10] models provide these pre-extracted features, significantly reducing the need for extensive computational resources during training.

Metrics This research project focuses on the NLQ benchmark. The Ego4D NLQ benchmark is framed as a temporal segment prediction task, where the model is provided with a video clip and a natural language query and tasked with predicting the temporal segment in which the answer is either visible or deducible. The metric “R@n, IoU= μ ” represents the percentage of language queries with at least one result whose Intersection over Union (IoU) with the ground truth exceeds μ in the top- n retrieved moments. Meanwhile, “mIoU” refers to the average IoU across all test samples. In our experiments, we set $n \in \{1, 5\}$ and $\mu \in \{0.3, 0.5\}$.

4.1. Training

In Tab. 1, we present the validation results from the Ego4D paper, obtained using VLSNet trained on SlowFast features. These results serve as a baseline for comparison in our analysis. We utilized two different types of word embeddings: Bert which generates contextual embeddings, where the vector representation of a word depends on its

surrounding words and GloVe which provides static embeddings, assigning the same vector to a word regardless of its context. When using GloVe, an RNN or Transformer is required as the final predictor to identify regions of interest in the videos. After analyzing different runs, we concluded that adjusting the learning rate does not lead to any significant improvement in the model's performance. Therefore, we are going to use a fixed learning rate for this phase ($\alpha = 0.0025$).

	IoU = 0.3		IoU = 0.5	
	r@1	r@5	r@1	r@5
SlowFast	5,45	10,74	3,12	6,63

Table 1. VLSNet trained on SlowFast features

4.1.1 Bert predictor

We now report the experiments done on EgoVLP features (Tab. 2) and on Omnivore (Tab. 3) changing some hyper parameters such as the dimension of the hidden size and the batch sizes using as text encoder Bert.

DM	BS	IoU = 0.3		IoU = 0.5	
		r@1	r@5	r@1	r@5
96	32	8,72	16,93	5,14	11,15
128	32	6,4	13,84	4,03	9,22
96	24	7,64	15,8	5,18	10,58

Table 2. VLSNet trained on EgoVLP (Bert)

DM	LR	IoU = 0.3		IoU = 0.5	
		r@1	r@5	r@1	r@5
96	32	6,61	14,25	3,74	9,01
96	16	6,89	13,58	3,92	8,62
128	16	7,63	13,86	4,08	8,60

Table 3. VLSNet trained on Omnivore (Bert)

From the experiments we can see that EgoVLP generally provides better recall scores than Omnivore across both IoU thresholds and most hyperparameter configurations. In Tab. 2 we can see that EgoVLP features demonstrate superior recall and benefit more from extended training epochs suggesting that they can capture more fine-grained temporal information with an extensive fine-tuning.

Omnivore features, as we can see in Tab. 3, show less variation with changes in batch size or hidden dimension compared to EgoVLP, on the other hand, they show robustness with less sensitivity to parameter changes but may require further adaptation to achieve comparable performance.

4.1.2 GloVe Predictor

We use a variation in the VLSNet architecture changing the Bert encoder with GloVe as textual encoder.

DM	LR	IoU = 0.3		IoU = 0.5	
		r@1	r@5	r@1	r@5
96	32	4,72	10,71	2,81	6,81
128	32	5,24	12,05	3,28	7,74
128	24	4,83	10,58	2,97	6,66

Table 4. VLSNet trained on EgoVLP (Glove)

In Tab. 6 and Tab. 7 we can notice that both EgoVLP and Omnivore with GloVe outperform SlowFast, but EgoVLP demonstrates significantly better results. EgoVLP offers the highest gains compared to SlowFast, making it the best-performing feature set among the three. Fine tuning for a higher number of epochs the best configuration with EgoVLP shows a consistent improvement in the performance, the opposite behaviour is shown for Omnivore under similar conditions.

4.2. FineTuning

Due to resource constraints, the final dataset was not fully completed. To address this, we implemented a function to remove all narratives for which no queries were generated. Additionally, it was not possible to retrieve all the features due to memory limits on Colab. Therefore, in this

DM	BS	IoU = 0.3		IoU = 0.5	
		r@1	r@5	r@1	r@5
96	32	4,39	9,27	1,99	5,34
96	16	6,12	13,53	3,41	8,72
128	32	4,21	10,79	2,27	6,09

Table 5. VLSNet trained on Omnivore (Glove)

DM	BS	IoU = 0.3		IoU = 0.5	
		r@1	r@5	r@1	r@5
96	32	4,72	10,71	2,81	6,81
128	32	5,24	12,05	3,28	7,74
128	24	4,83	10,58	2,97	6,66

Table 6. VLSNet trained on EGOVLP (Glove)

DM	BS	IoU = 0.3		IoU = 0.5	
		r@1	r@5	r@1	r@5
96	32	4,39	9,27	1,99	5,34
96	16	6,12	13,53	3,41	8,72
128	32	4,21	10,79	2,27	6,09

Table 7. VLSNet trained on Omnivore (Glove)

section, we analyzed only EGOVLP with the two different encoders presented: BERT and GloVe.

DM	BS	IoU = 0.3		IoU = 0.5	
		r@1	r@5	r@1	r@5
96	32	1.32	0.67	3.15	1.45
128	32	1,63	0,86	4,47	2,19
96	24	1,55	0,85	3,48	1,68

Table 8. VLSNet trained on EgoVLP (Bert) after finetuning

DM	BS	IoU = 0.3		IoU = 0.5	
		r@1	r@5	r@1	r@5
96	32	8,03	4,93	15,59	10,61
128	32	8,85	5,45	16,65	10,94

Table 9. VLSNet trained on EgoVLP (Glove) after finetuning

The contrasting performance between BERT Tab. 8 and GloVeTab. 9 when using LLM-generated data can be attributed to their fundamental architectural differences and how they process language. BERT’s worse performance is likely because it’s a sophisticated contextual model that’s highly attuned to natural language patterns, making it more sensitive to the subtle artificialities in LLM-generated text. Its bidirectional nature and deep contextual understanding mean it can detect when the linguistic patterns don’t quite match what it learned during its original pretraining. In contrast, GloVe’s better performance stems from its simpler approach based on word co-occurrence statistics. Since GloVe doesn’t try to understand deep contextual relationships but instead focuses on basic word-level associations, it’s less affected by the potential unnaturalness or oversimplification in synthetic data. The word relationships that GloVe captures are likely preserved adequately in LLM-generated text, allowing it to maintain its performance even with synthetic data. This suggests that while LLMs can generate grammatically correct and meaningful text, they might not fully replicate the subtle linguistic patterns that more sophisticated models like BERT are trained to recognize.

5. Conclusion

In conclusion, after becoming familiar with the Ego4D dataset and analyzing its Natural Language Queries (NLQ) annotations, we trained VSLNet with its two variants, VSLBase and VSLNet. We focused on training them first with features extracted using Omnivore, and then with EgoVLP. In the end, we implemented an automatic query generation system using LLMs with the support of Gemma. The goal was to use a large language model to generate query-answer pairs from a set of randomly sampled consecutive narrations.

Comparing the results, we obtained better outcomes using the BERT encoder over the VSLNet network trained on SlowFast features. However, using GloVe, we achieved better results only once, with a hidden size dimensionality of 86 and a batch size of 16. Regarding fine-tuning, the results were opposite: the network trained with the BERT encoder performed poorly, while GloVe surprisingly outperformed expectations.

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